Problem 5.1

A principal of \$2,000 is deposited on June 17 and withdrawn on September 10 of the same year, at a simple interest rate of 8% per year. Find the interest earned using:

(a) exact simple interest,

(b) ordinary simple interest,

(c) Banker's Rule.

We use the simple interest formula:

$$I = Prt,$$

where $P = 2000$, $r = 0.08$, and t is the time expressed in years.

1. Number of Days

From June 17 to September 10:

$$13 + 31 + 31 + 10 = 85 \text{ days.}$$

For ordinary simple interest, 30-day months are used:

$$13 + 30 + 30 + 10 = 83 \text{ days.}$$

(a) Exact Simple Interest

For exact simple interest, the exact number of days is used together with a 365-day year:

$$I = 2000(0.08)\left(\frac{85}{365}\right).$$

Therefore,

$$I \approx 37.26.$$

(b) Ordinary Simple Interest

For ordinary simple interest, the approximate time of 83 days is used together with a 365-day year:

$$I = 2000(0.08)\left(\frac{83}{365}\right).$$

Thus,

$$I \approx 36.38.$$

(c) Banker's Rule

Under Banker's Rule, the exact 85 days are used, but with a 360-day commercial year:

$$I = 2000(0.08)\left(\frac{85}{360}\right).$$

So,

$$I \approx 37.78.$$